

EXERCISES

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UA2. DATA ACCESS

EXERCISE 1

The text file *supermarket_sales.txt* contains data relating to historical sales of a supermarket from January 2019 to March 2019. The organization of this file is described in the following table:

Field	Columns	Description
<i>Invoice_ID</i>	1 - 11	Computer generated sales slip invoice identification number
<i>Branch</i>	12	Branch of supermarket
<i>City</i>	13 - 21	Location of supermarket
<i>Customer_Type</i>	22 - 27	Type of customers
<i>Gender</i>	28 - 33	Gender of customer
<i>Product_Line</i>	34 - 55	General item categorization group
<i>Unit_Price</i>	56 - 60	Price of each product in \$
<i>Quantity</i>	61 - 62	Number of products purchased by customer
<i>Tax_5</i>	63 - 69	5% tax fee for customer buying
<i>Total</i>	70 - 77	Total price including tax
<i>Date</i>	78 - 86	Date of purchase
<i>Time</i>	87 - 91	Purchase time
<i>Payment</i>	92 - 102	Payment used by customer for purchase
<i>Rating</i>	103 - 105	Customer satisfaction rating on their overall shopping experience

- a) Use INPUT column to create a data set named *smsales1* based on the text file *supermarket_sales.txt*, such that:
- The *smsales1* data set contains only the invoice ID, the branch of supermarket, the total price including tax and the date of purchase;
 - The *smsales1* data set contains the text file records from 100 to 200;
 - Verify in the Log window, the number of observations and the number of variables in the *smsales1* data set.

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- b)** Create a new data set, *smsales2*, using a data step, such that:
- contains all the observations in *supermarket_sales.txt* file;
 - Use formatted INPUT to read the invoice ID, the branch of supermarket, the location of supermarket, the general item categorization group, the total price including tax and the date of purchase;
 - Format the *Date* variable so as to have the appearance 05JAN2019.
- c)** Considering the data step from the previous paragraph, make the necessary changes so that the *smsales2* data set:
- only includes the first 300 observations;
 - only includes transactions with a total price including tax exceeding 200.
- d)** Considering the data step from paragraph b), make the necessary changes so that the *smsales2* data set only includes transactions for product lines categorized as 'Health and beauty' and product lines categorized as 'Home and lifestyle'.

EXERCISE 2

See the *employees.txt* file, which contains information regarding the employees of a particular company. The fields of this text file are, in this order:

Variable	Description	Type
Id	Identification	Character
Surname	Last name	Character
Name	Name	Character
City	City	Character
Sex	Gender	'F' or 'M'
Cod	Department code	Character
Salary	Salary	Numeric
Birth	Date of birth	Date

- a)** Create a data set named *employees1* that includes all variables and all observations of the text file. Analysing the Log window, can you tell how many observations exist in the data set that was created?

- b) Considering the program from the previous paragraph, adapt it so that the *employees1* data set has the same column names as in the table above.
- c) Considering the program from the literal a), adapt it so that the *employees1* data set includes only those employees with a salary greater than 30,000.
- d) How many male employees are there?

EXERCISE 3

Consider the following table of data:

Invoice_ID	Date	Customer_ID	Gender	Product_ID	Quantity	Unit_Price	Total
1001	27MAR23	C062	F	P023	2	60	120
1002	05JUN23	C052	F	P005	2	100	200
1003	04JUL23	C002	M	P011	3	50	150
1004	27SEP23	C187	M	P020	1	200	200
1005	09JAN24	C229	M	P033	2	80	160
1006	13FEB24	C034	F	P104	1	150	150

- a) Create a data set named *invoices1* with these data.
- b) Adapt the previous data step in order to create a new data set named *invoices2*, where the date appears as 27/03/2023 and only contains transactions from male customers with a total purchase amount lower than or equal to 160.
- c) Create a data set named *invoices3* that includes all transactions with a total purchase amount between 120 and 150, inclusive.

EXERCISE 4

See the *adidas_us_sales.xlsx* file, which contains sales data of Adidas in United States.

- a) Create a data set named *adidas_us_sales* from this Excel file using the *Import Wizard*.
- b) Similarly, create a data set named *adidas_us_sales* from this Excel file using PROC IMPORT.

EXERCISE 5

See the *walmart_sales.txt* file, which contains sales data of Walmart and includes:

- The store number;
 - The department number;
 - The week;
 - Sales for the given department in the given store;
 - Whether the week is a special holiday week.
- a) Create a dataset named “walmart_sales” (not temporary) providing a name of your choice for the variables.
- b) Create a new dataset based on the first one. In the new one you should create a new variable named “Sales_Per_Day” that represents the average sales per day for each entry.

EXERCISE 6

Identify the mistake(s) in the following SAS code:

```
data new-data;
    infile prob4data.txt;
    input x1 x2
    y1 = 3(x1) + 2(x2);
    y2 = x1 / x2;
    new_variable_from_x1_and_x2 = x1 + x2 - 37;
run;
```